CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (EC/ME/CL/CE/IT/EE)

May 2012

MA101 Engineering Mathematics-I

Date: 10/05/2012, Thursday Time: 10:00am to 01:00pm Maximum Marks: 70

Instruction:
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 State wether the following statements are true or false [07]

(a) Geometric series $\sum\limits_{n=0}^{\infty} ar^n$ is convergent for $|r| < 1$.

(b) According to Cauchy p-test, $\sum \frac{1}{n^p}$ is divergent for $p \leq 1$.

(c) Complex number $4 + i\,3$ can be represent as $5\, e^{i \tan^{-1}\frac{3}{4}}$.

(d) $\tan ix = i \tanh x$.

(e) If determinant of a matrix is zero then inverse of the matrix does not exist.

(f) If rank of 3×4 matrix is 3 then nullity of the matrix is 1.

(g) Homogeneous system is always consistant.

Q - 2(a) Discuss the convergence of $\sum\limits_{n=2}^{\infty} \frac{1}{n\sqrt{n^2-1}}$ [04]

(b) Prove that $(1+i)^n + (1-i)^n = 2^{\frac{n}{2}+1} \cos\left(\frac{n\pi}{4}\right)$. [04]

(c) Using Gauss-Jordan method find inverse of $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$. [04]

(d) Find value of $\sum\limits_{n=0}^{\infty} \left(\frac{8}{11}\right)^n$ [02]

OR

Q - 2(a) Find radius of convergence for $\sum\limits_{n=1}^{\infty} \left(1+\frac{1}{n}\right)^n x^n$. [04]

(b) Find forth root of $1+i$. [04]

(c) Solve by Gauss elimination method: $x + 2y + z = 4, 2x + y = 3, x + z = 2$. [04]

(d) Find row-echelon form of $\begin{bmatrix} 2 & 3 & 1 \\ 1 & 1 & 0 \\ 5 & 0 & -2 \end{bmatrix}$. [02]

Q - 3 (a) Determine the value of $\sum\limits_{n=2}^{\infty} \frac{1}{n(n-1)}$. [04]

(b) If $x_r = \cos\frac{\pi}{2^r} + i \sin\frac{\pi}{2^r}$, prove that $\lim\limits_{n\to\infty} x_1 x_2 x_3 \cdots x_n = -1$. [04]

(c) For what value of k, the equations $x + 2y + z = 3$, $x + y + z = k$ and $3x + y + 3z = k^2$ are consistent? [04]

(d) Solve: $x + 2y - z = 0,\ 2x - y = 0$ and $y - z = 0$. [02]

OR

Q - 3(a) Discuss the convergence of $\sum_{n=0}^{\infty} \frac{n!}{1000^n}$. Give the name of test which you applied. [04]

(b) Find Modulus and argument of i^{1-i}. [04]

(c) Find rank and nullity of $\begin{bmatrix} 2 & 1 & 6 & -4 \\ 1 & 0 & 3 & -2 \end{bmatrix}$. [04]

(d) Simplify: $(\cos 3\theta + i \sin 3\theta)^2 (\cos 2\theta - i \sin 2\theta)^3$. [02]

SECTION - II

Q - 4 State wether the following statements are true or false [07]

(a) L'Hospital rule can be directly applicable for $\infty - \infty$ type undetermined form.

(b) Cauchy mean value theorem is particular case of Lagrange mean value theorem.

(c) Taylor series expansion of a function is Maclaurin's expansion of the function at 0.

(d) According to Euler's theorem, if $u = cf(x, y)$is homogeneous function with degree n, then $xu_x + yu_y = u$.

(e) We can not find total derivative of function of more variable.

(f) Normal line and tangent plane may not be perpendicular at some point on a curve.

(g) Local maxima is always smaller than Global Maxima of function.

Q - 5(a) Find n^{th}derivative of $y = \frac{2x-1}{(x-2)(x+1)}$. [04]

(b) If $u = \tan^{-1}\left(\frac{y}{x}\right)$, then prove that $u_{xy} = u_{yx}$. [04]

(c) If $x = r\cos\theta,\ y = r\sin\theta$, find Jacobians $\frac{\partial(x, y)}{\partial(r, \theta)}$ and $\frac{\partial(r, \theta)}{\partial(x, y)}$. [04]

(d) Find Maclaurin's series expansion of $f(x) = e^x$. [02]

OR

Q - 5(a) Verify Lagrange's Mean value theorem for $f(x) = \tan^{-1}x,\ x \in [-1, 1]$ and find $c \in [-1, 1]$. [04]

(b) If $xyz = c$, prove that $\left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x \left(\frac{\partial y}{\partial x}\right)_z = -1$. [04]

(c) Examine the function $f(x, y) = x^3 + y^3 - 3axy,\ (a > 0)$ for maxima and minima. [04]

(d) If $u = x^2 + y^2$, find $xu_x + yu_y$. [02]

Q - 6(a) Find Taylor series expansion of $f(x) = \sin x$ at point $\frac{\pi}{4}$. [04]

(b) If $u = \log\left(\frac{2x}{y}\right)$, prove that $xu_x + yu_y = 0$ and $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 0$. [04]

(c) Find approximate value of $\sqrt[3]{1021}\,\sqrt{27}$. [04]

(d) Find equation of tangent plane to the surface $z = x^2 + y^2 - 2$ at $(1, 2, 1)$. [02]

OR

Q - 6 (a) If $\log y = m\sin^{-1}x$, prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$. [04]

(b) Verify Euler's theorem for $u = ax^2 + 2hxy + by^2$. [04]

(c) Find the maximum value of $V(x, y, z) = xyz$ subject to the constraint $2x + 2y + 2z = 108$. [04]

(d) Fimd n^{th}derivative of $y = \cos(2x - 7)$. [02]

****